EXECUTIVE SUMMARY | MANAGEMENT REPORT | ILLUSTRATIVE BUSINESS

Studio North: is growth becoming more valuable?

Customer acquisition, monthly renewal and contribution economics | 24 monthly acquisition cohorts, 2024-01 through 2025-12; data cutoff 2026-01-01

THE BUSINESS AND THE DECISION

Studio North sells a monthly creative-work membership to independent professionals. Templates support immediate tasks; guided learning builds skills. Members choose standard or premium plans, may buy add-ons, and decide monthly whether to renew. Search and partner workshops acquire customers seeking time savings or stronger skills.

Management needs to know whether successive acquisition cohorts create more durable contribution at a cost the company can recover quickly enough to fund growth.

Customer economics are strengthening: lower blended CAC, higher contribution and more cash recovered over six months. Retention also improves within each channel–need group after excluding both experimental arms. Short campaign tests do not establish the return from a larger budget.

ACQUISITION EFFICIENCY
$108 → $81
blended CAC per acquired customer

CUSTOMER VALUE
$107 → $130
six-month contribution per customer

CASH RECOVERY
-$1 → $49
six-month cash after blended CAC

Matched-age comparison: January–June 2024 versus January–June 2025; age 5, six observed receipts; acquired-customer weights. Values above are rounded to dollars; the evidence table reports cents. Differences use unrounded values.

WHAT THE EVIDENCE ESTABLISHES

Table 1. Matched-age evidence on acquisition efficiency, customer value and cash recovery.

Evidence	Earlier	Later	Interpretation
Sixth-month retained share	52.2%	60.9%	Observed at age index 5
Blended CAC	$107.67	$80.53	Tracked spending / all acquired people
Six-month contribution / customer	$106.52	$129.98	After recorded variable costs
Six-month cash after acquisition	-$1.16	$49.46	Contribution less blended CAC
Ex-test retained share	52.2%	61.6%	Both experimental arms excluded
Ex-test CAC / cash	$107.67 / -$1.16	$85.31 / $45.80	Same joining windows; both test arms excluded

The later window includes 98 experimental buyers and $650 of test spending; the earlier window has none. Sensitivity rows exclude both arms and their receipts. Cash improves by $46.95 excluding tests, versus $50.62 including them. Differences are computed from unrounded values.

Initial need is informative: build skill customers show $40 more six-receipt contribution and 23 percentage points higher retention than save time customers; variation within each group remains substantial.

RECOMMENDED MANAGEMENT RESPONSE

Action. Marketing and Analytics should verify that controls stayed protected before using later outcomes; otherwise run a longer test. Finance should monitor the ex-test $45.80 six-month cash benchmark, not use it as an automatic causal gate. Product should investigate, using events and feedback, what changed for the later cohorts. Incremental contribution after media per assigned prospect: search / save time: $1.91 over 31 days (95% interval $0.03 to $3.78); partner / build skill: -$0.34 over 30 days (95% interval -$2.33 to $1.65). 1 of 2 intervals include zero; Search is encouraging; neither short window establishes lifetime returns.

CONDITIONS FOR REVISITING THE DECISION

Revisit the growth plan if later cohorts' six-month cash recovery falls toward the earlier window, if a valid longer campaign comparison fails to cover media cost, or if young-cohort outcomes fall materially below forecast. Historical blended CAC does not establish the return from another advertising dollar.

Studio North and the customer experience

The membership business, customer needs and possible sources of value and friction.

The membership experience behind the economics

Studio North serves independent professionals seeking faster creative work or stronger skills.

Templates support immediate work; guided lessons build capabilities over repeated use. A monthly membership is valuable when those benefits recur often enough to justify the next payment. Optional workshops and add-ons expand spending but do not by themselves establish durable membership value.

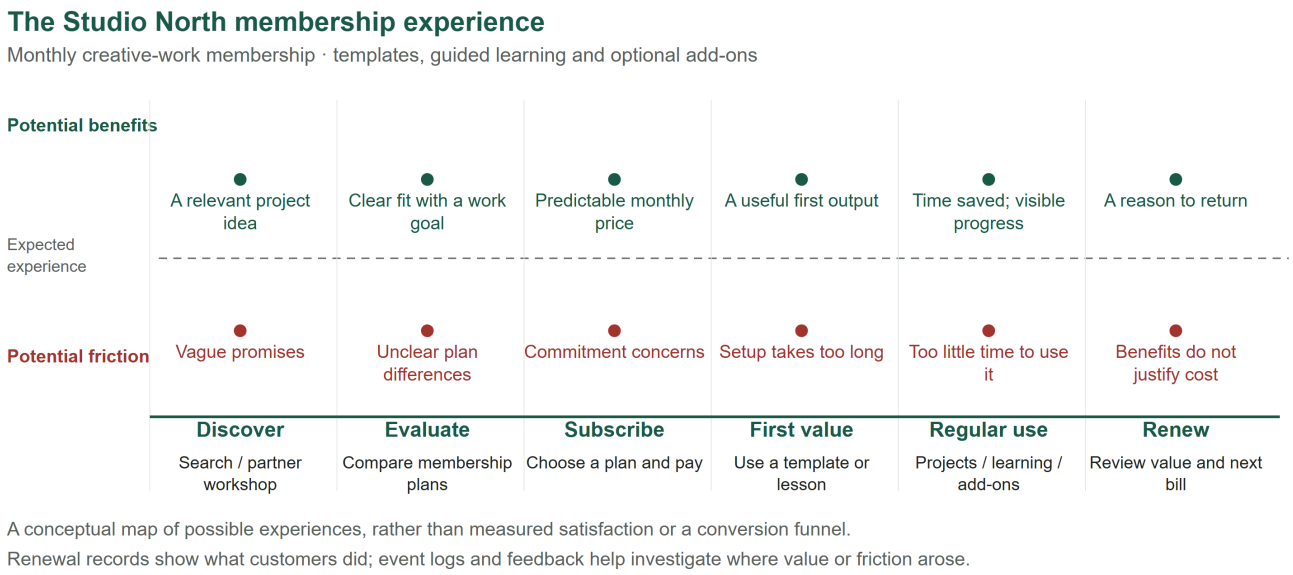


Figure 1. Conceptual. A broad conceptual journey through the Studio North membership. Benefits and friction points are possible experiences, not measured sentiment or population conversion rates.

POTENTIAL SOURCES OF RENEWAL FRICTION

An unclear plan choice can delay subscription. A difficult first task can prevent a member from experiencing value. Later, a lack of time or an unmet learning goal may weaken the reason to renew. These are business hypotheses for investigation; the source records do not establish their causes.

EVIDENCE AVAILABLE TO THE MANAGEMENT TEAM

The CRM records stated needs and baseline characteristics. Product events show activity, transactions show spending, renewal records show continuation, and feedback captures selected perceptions. Together, these sources support a more informative diagnosis than a renewal rate alone. Tables 12–19 classify all eight sources using the Universal Customer View. The appendix documents parameters, validation and reproducibility.

> All records describe an illustrative business and are synthetic. Recorded variable costs and acquisition spending exclude fixed operating costs, financing and taxes; this analysis does not establish company profitability.

Observed history and the forecast horizon

How a renewal model supports a common comparison of mature and newly acquired cohorts.

Extrapolation makes young cohorts comparable

A common twelve-receipt horizon combines observed contribution with an explicit forecast of the unobserved remainder.

WHAT THE MODEL CONTRIBUTES

Recent cohorts have not yet had the opportunity to demonstrate a full year of renewals. The retention model estimates how continuation varies across customers, acquisition channels, stated needs and joining months. It uses those patterns to forecast the remaining renewal path. A revenue model then translates retained membership into expected contribution, which is discounted and compared with acquisition cost.

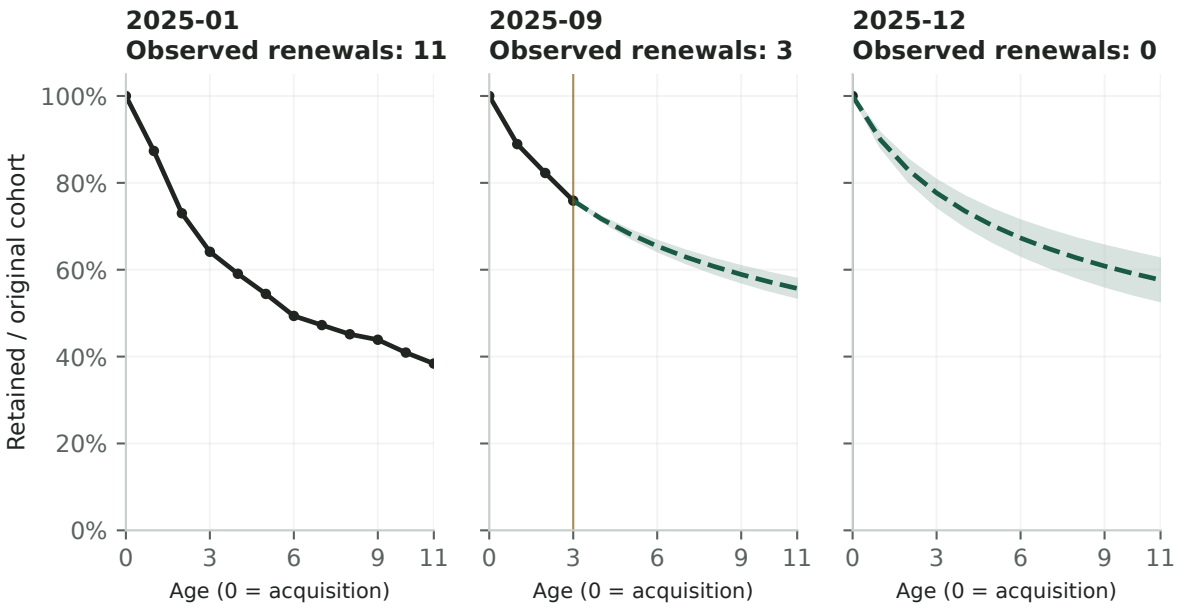


Figure 2. Observed and forecast. Observed and forecast retention for cohorts with complete, intermediate and minimal renewal histories. Solid points are observed; dashed continuations are forecasts. Examples are selected by history length, not outcome.

Saved model output: `forecast_curve.csv`. Original cohort denominators are preserved. Shading shows the 5th–95th posterior parameter range; the vertical marker identifies the observation boundary.

THE INFORMATION BEHIND EACH FORECAST

The mature cohort supplies a fully observed reference. The intermediate cohort combines its own renewal history with evidence from other cohorts. The newest cohort has only its acquisition month and therefore **zero observed renewal decisions**; its forecast depends primarily on comparable customers and model assumptions.

> Extrapolation supports an earlier assessment of cohort value; it does not turn future outcomes into observed facts. These bands describe parameter uncertainty in expected paths, excluding customer-level process variation and changes in the business. The model specification, historical prediction checks and valuation assumptions appear later in the report.

Observed cohort economics

Retention, contribution and acquisition-cost recovery at comparable customer ages.

Customer retention at comparable cohort ages

A joining cohort contains customers whose first successful payment occurred in the same month. Future ages remain unobserved.

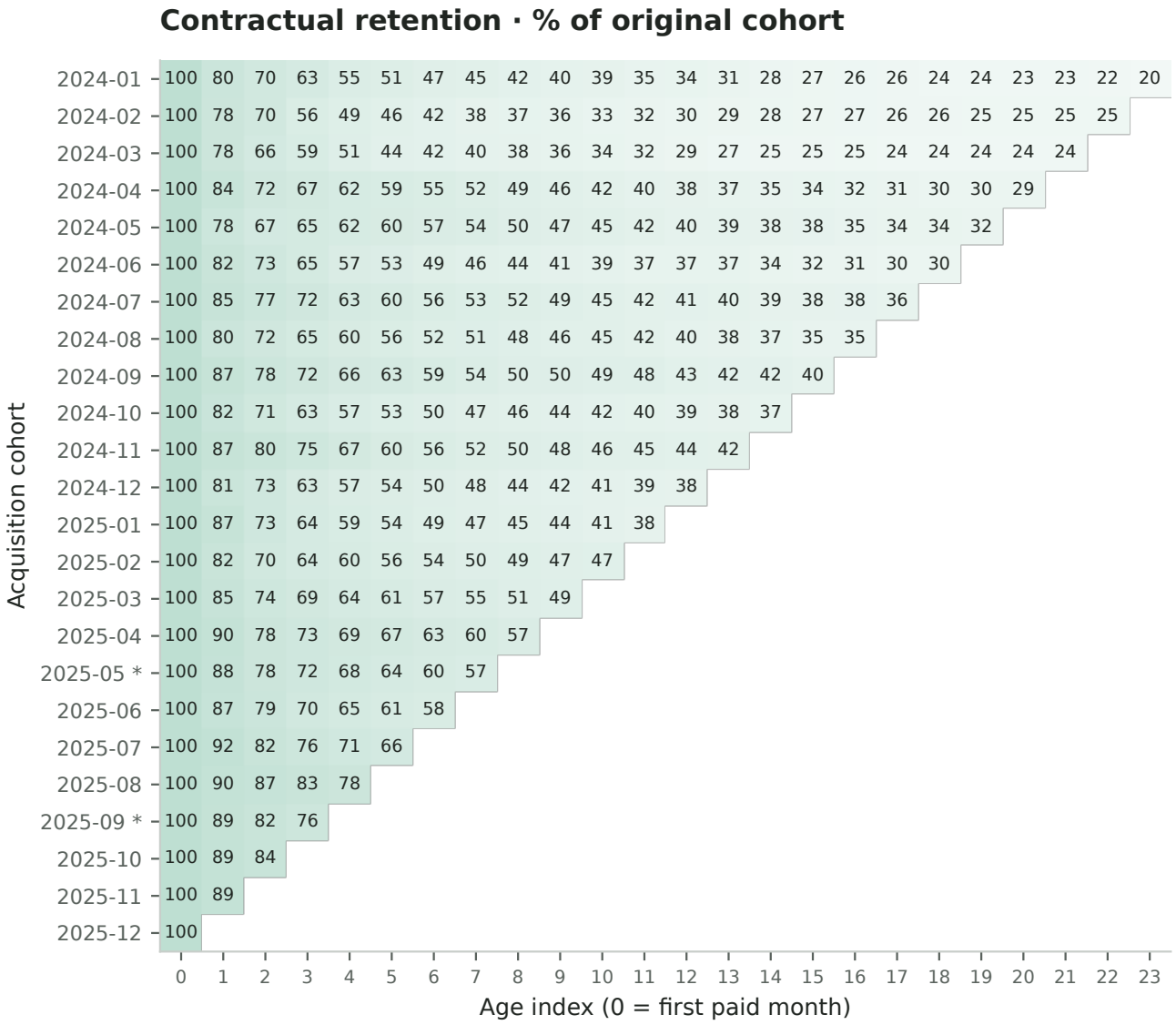


Figure 3. Observed. Customer retention for all 24 monthly acquisition cohorts and every observed age. Cells are percentages of the original cohort; blanks are unobserved. * marks test months, including buyers from both arms.

Saved SQL output: `cohort_totals.csv`. The full age range is 0–23; age 0 is the first paid month. Rows follow individual cohorts; columns compare successive cohorts at the same age. No missing future month is assigned zero retention.

COHORT AGE AND CALENDAR TIME

Every cohort starts at 100% because acquisition defines its original denominator. Newer cohorts have shorter histories. Their shorter lines and blank cells do not establish shorter customer lifetimes. Figure 5 shows the same full cohort sequence as curves; Figure 7 recombines cohorts in calendar time.

An improving business should acquire cohorts that retain more customers at comparable ages, rather than relying only on a larger inflow of new customers.

Contribution and acquisition-cost recovery

All 24 cohorts | Cash balance uses recorded contribution, tracked acquisition cost and the original acquired-customer denominator.

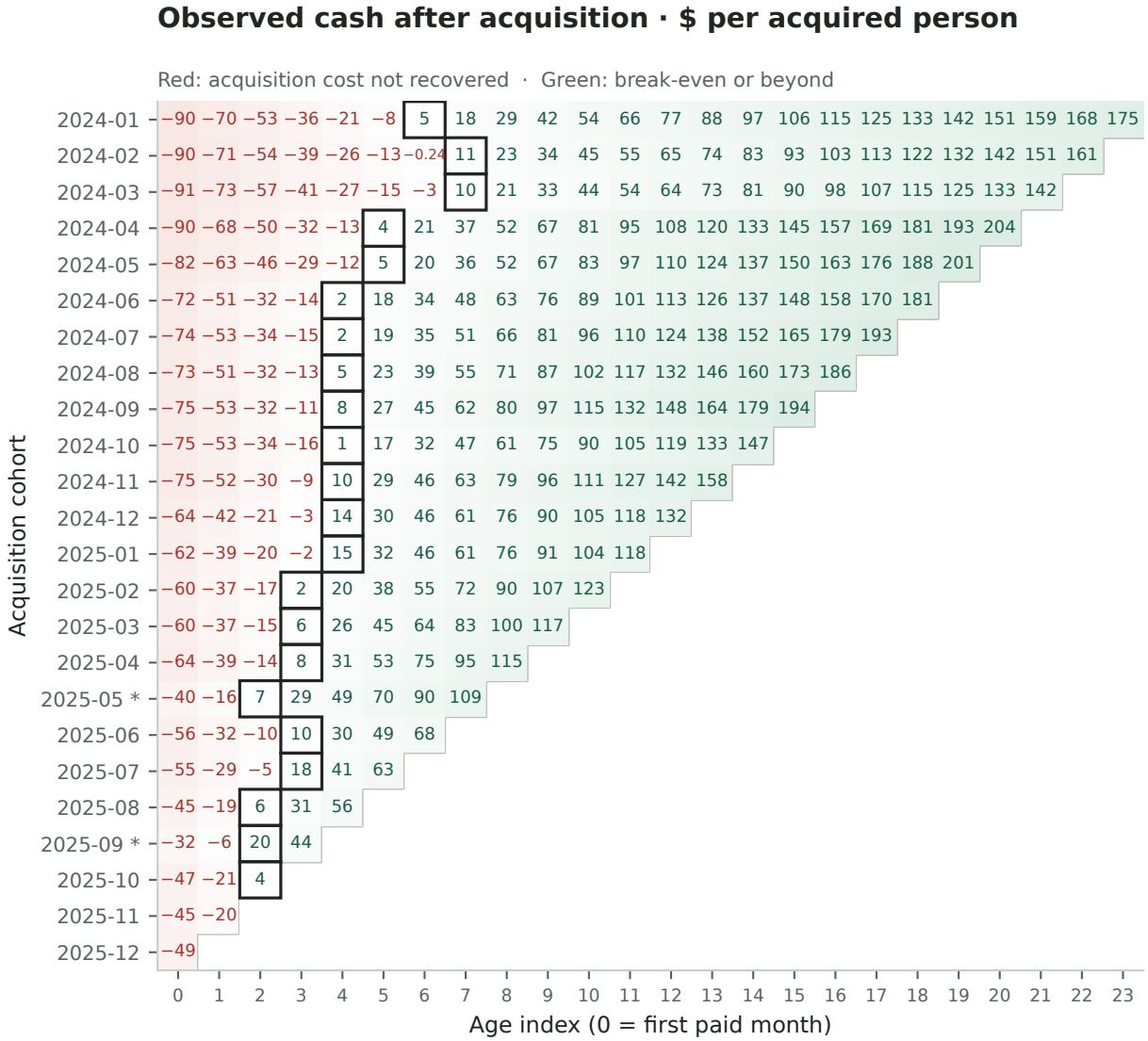


Figure 4. Observed. Cumulative cash after acquisition for every joining cohort, in dollars per original acquired customer. Red numerals indicate contribution below CAC; green numerals indicate break-even or beyond. An outline marks the first observed recovery of CAC. * marks test months, including both arms.

Saved SQL output: `cohort_cash_economics.csv`. Each cell equals cumulative net revenue less incurred variable costs and tracked acquisition spending, divided by original cohort size. The blank region is unobserved. This is undiscounted cash, not total company profit or discounted LTV.

ACQUISITION EFFICIENCY AND DURABILITY MUST MOVE TOGETHER

At age 5, six-month cash after acquisition changes from -$1.16 to $49.46 per original customer. The comparison uses January–June 2024 and the same joining months of 2025, weighted by acquired cohort size. Table 1 reports the matching retention, contribution and blended-CAC evidence. These are descriptive comparisons, not campaign effects.

> A new cohort may recover acquisition cost sooner because it costs less to acquire, generates more contribution per retained member, retains more members, or combines these changes. The retention and cash triangles together distinguish these sources of change.

Cohort trajectories

The relationship between member retention and revenue durability across the full sequence.

Retention and revenue across the cohort sequence

Every monthly acquisition cohort is shown. Shared axes and one chronological color scale keep the comparisons aligned.

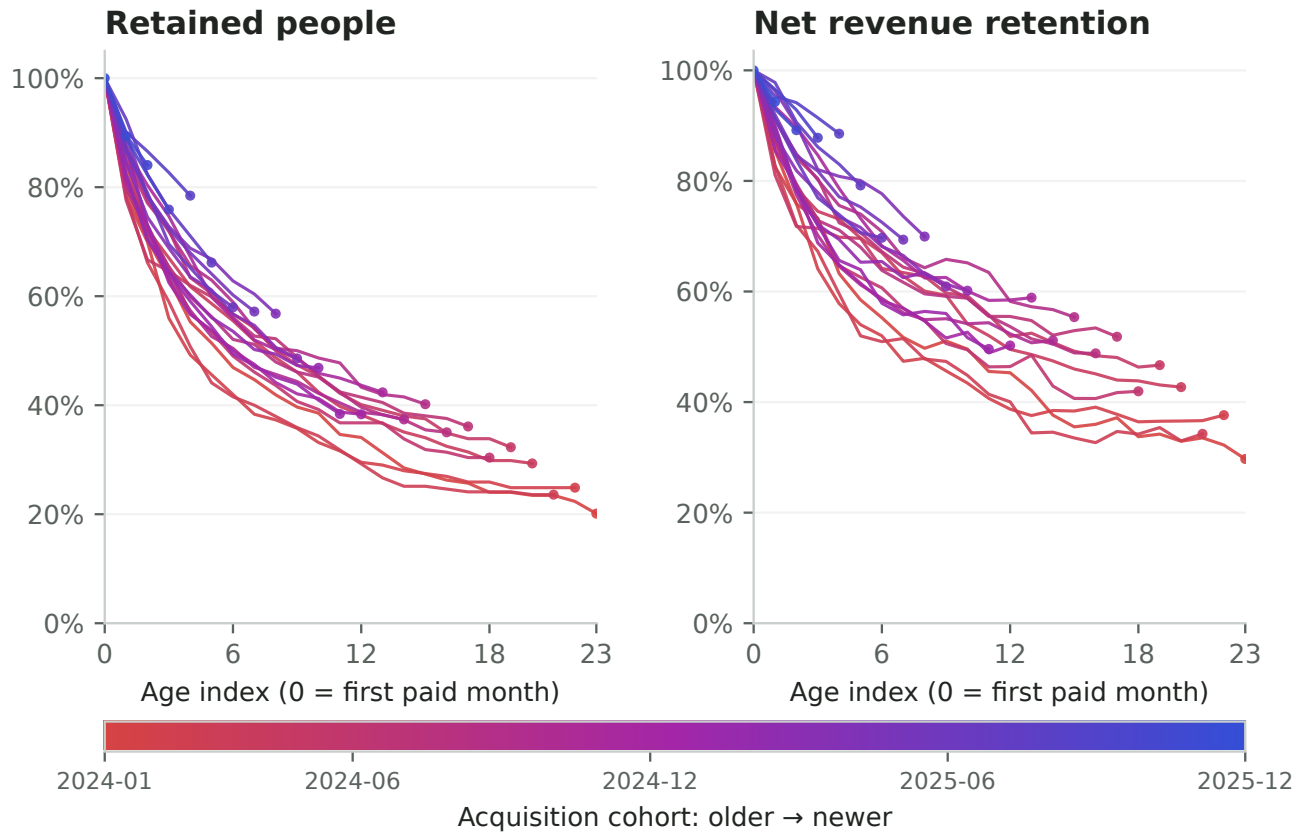


Figure 5. Observed. All 24 customer-retention and revenue-retention curves. Older cohorts are red and newer cohorts are blue; each line ends at its last observed month.

Retained share uses original cohort membership; revenue retention uses each cohort's first-period revenue. Revenue can change because of the number of renewing members, their plan mix or add-on spending. Product activity is not a substitute for a renewal record.

COHORT TRAJECTORIES AT COMPARABLE AGES

The vertical position of differently colored curves at the *same* age supports an aligned comparison. Later cohorts lying above earlier cohorts suggest improving retention; they need not dominate at every month. The chronological color key replaces an unwieldy 24-item legend. A curve with only its acquisition point has no observed renewal history.

CUSTOMER COUNTS AND REVENUE ANSWER DIFFERENT QUESTIONS

Higher revenue retention can coexist with customer losses when remaining members upgrade or buy add-ons. A retention improvement should therefore be checked against contribution and acquisition recovery in Figure 4, and against comparable customer groups in Figure 6.

The trajectory matters as well as the cutoff. A stronger first month does not establish longer customer life, while a customer-count decline can coexist with more spending by the members who remain.

Customer segmentation

Baseline customer differences, within-segment progress and implications for the operating plan.

Customer differences and within-segment progress

Observed renewal trajectories by acquisition channel and stated need; each panel uses the same axes and all 24 cohorts.

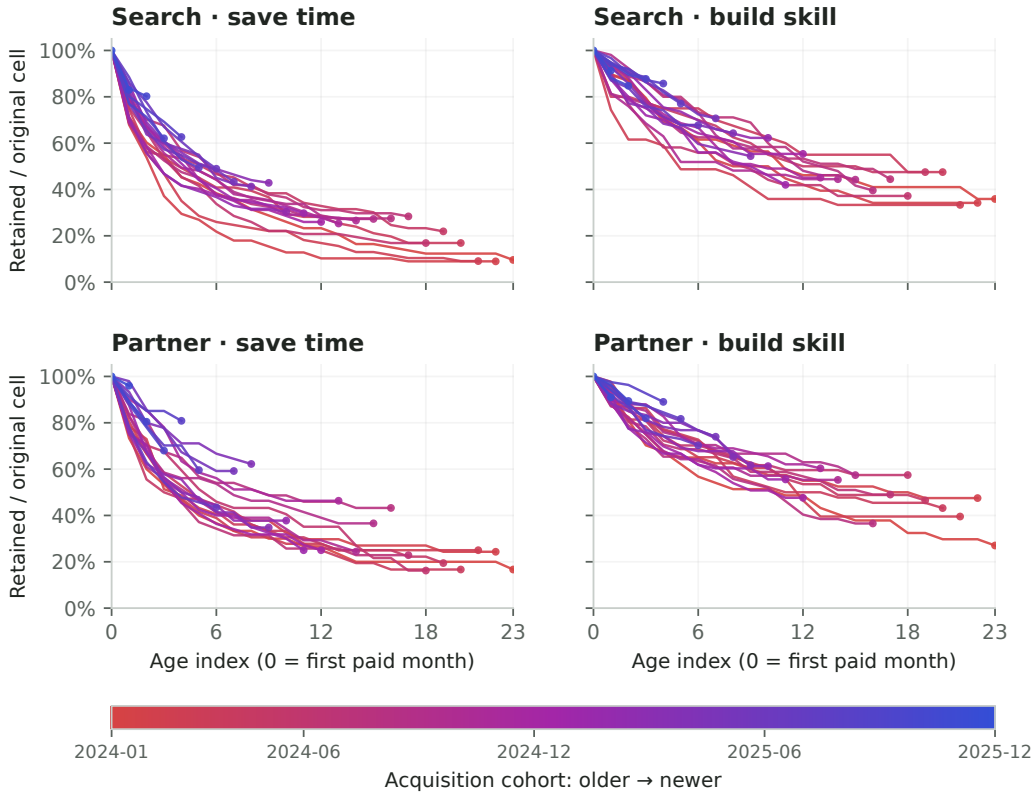


Figure 6. Descriptive comparison. The full cohort sequence within each channel-by-need group. The color key identifies acquisition month. Paid channels are shown; organic test-control cells are omitted.

Saved SQL output: `cohort_age_summary.csv`. Curves divide retained members by the original cohort–channel–need cell. No missing future observations are filled with zeros.

WITHIN-GROUP DIAGNOSIS

Excluding both experimental arms, age-5 paid retention is 52.2% for January–June 2024 versus 61.6% for the same 2025 joining months. Fixing the earlier channel-by-need mix gives 52.2% versus 60.2%; 4 of 4 paid groups improve. This removes changes in these groups' shares, not all customer differences or confounding. The curves above retain the full observed population; the table excludes both experimental arms.

Table 2. Sixth-month paid retention by original acquisition channel and baseline stated need; the same joining months one year apart.

Original channel / need	Jan–Jun 2024	Jan–Jun 2025	Change
Search / save time	39.0%	50.3%	+11.3 pp
Search / build skill	64.4%	69.9%	+5.5 pp
Partner / save time	45.3%	53.1%	+7.9 pp
Partner / build skill	69.7%	74.6%	+4.9 pp

Both experimental arms are excluded from this table and the fixed-mix calculation. The curves above include test buyers. Differences use unrounded values; the whole-company ex-test sensitivity appears in Table 1.

Segmentation explains part of the economic variation

Baseline characteristics define customer groups; their observed outcomes establish how useful those groups are for this decision.

ECONOMIC DIFFERENCES ACROSS CUSTOMER GROUPS

Customers whose initial stated need is build skill produce $39.59 more contribution over six receipts than those stating save time, on average. Their sixth-month retained share differs by 23.1 percentage points. This identifies an economically relevant customer distinction, not proof that changing a customer's need or acquisition channel would cause the same outcome.

Table 3. Six-receipt economics for the same mature paid customers, excluding both experimental arms. SD is the standard deviation of individual contribution within the stated-need group.

Initial stated need	Customers	Retained at age 5	Six-receipt contribution	Within-group SD
build skill	2,002	70.4%	$142.87	$66.79
save time	2,276	47.2%	$103.28	$64.54

Every comparison above uses the same mature paid customers, including zero contribution after churn. Differences use unrounded values. Both experimental arms are excluded. Table 20 compares four two-group segmentation strategies; cohort centering separates shared joining-month levels from within-group variation.

IMPLICATIONS FOR THE OPERATING PLAN

The more durable group supports a longer relationship and a different cash-recovery profile. The less durable group may still be attractive at an appropriate acquisition cost, with a service proposition suited to its shorter use cycle. Product and Marketing can distinguish immediate task completion from ongoing capability development when designing onboarding and renewal communications.

> Segmentation is a useful summary of market heterogeneity, not a claim that members of a group are identical. These comparisons identify where to investigate or test a decision; they do not estimate the effect of moving customers between groups.

Company economics

How successive cohorts accumulate into an active membership base and operating contribution.

Retained cohorts determine how growth accumulates

The company is the sum of its acquisition cohorts at different ages, not the latest cohort scaled up.

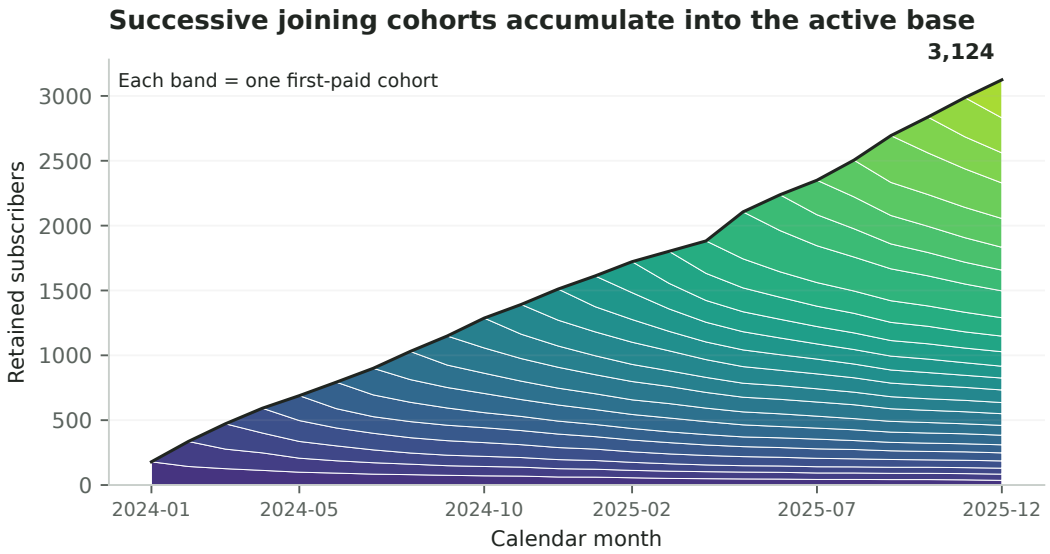


Figure 7. Observed. The company's retained membership base as the sum of all 24 acquisition cohorts in calendar time.

Saved SQL output: `company_month.csv`. Each layer is a joining cohort's active customers at the displayed calendar month. Layer totals are reconciled to independently saved company totals before plotting.

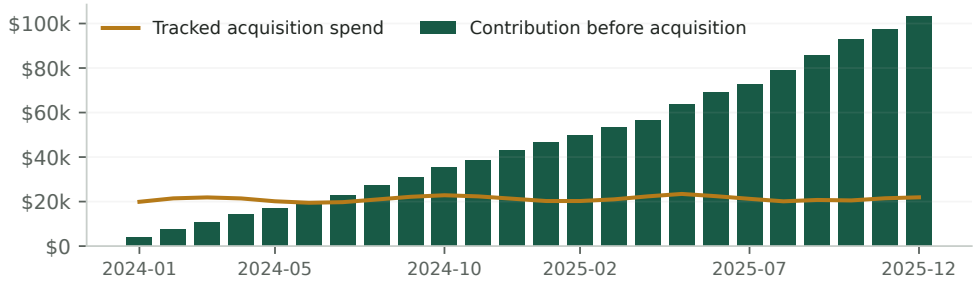


Figure 8. Observed. Monthly contribution before acquisition compared with tracked acquisition spending.

SCALE AND CASH RECOVERY ARE SEPARATE TESTS

New acquisition adds customers immediately. Renewals preserve older layers and their contribution, reducing the replacement acquisition required to sustain scale. The lower chart compares recorded contribution with acquisition spending in each calendar month; it is not cumulative retained customers or a profit statement. In 2025-12, recorded contribution was $103,343 and acquisition spending was $21,915, leaving $81,428 before fixed costs, financing and taxes. This period result should be read alongside the age-aligned cohort comparison, rather than used as a substitute for it.

OPERATING IMPLICATION

The operating review benefits from both the common-age economics of each joining cohort and the company's accumulated contribution. A favorable acquisition metric can coexist with slower cash recovery when renewals weaken. A growing customer base can also conceal deterioration if management replaces departing customers with increasingly costly new ones.

Cash payback is undiscounted: the first age at which cumulative contribution covers the acquisition investment. The LTV calculation that follows discounts expected monthly contribution and uses a common, explicit horizon.

Customer value forecasts

The renewal model, contribution valuation and the assumptions behind immature-cohort LTV.

Monthly renewal and the valuation model

Hierarchical shifted-Beta-Geometric retention + Gamma gross revenue; NumPyro NUTS

HOW THE MODEL REPRESENTS THE BUSINESS

Members differ in their underlying propensity to renew. The model learns how that distribution varies with initial need, acquisition channel and joining cohort. A cohort's average churn can decline as its more durable members remain; this need not mean each individual becomes more loyal. Completed renewals provide evidence, while pending and future decisions remain unobserved.

$$\text{net LTV}_H = \sum_{a=0}^{H-1} \frac{\text{E}[\text{contribution}_a \mid \text{observed history}]}{(1+d)^{a+1}} - \text{CAC}\,.$$

A separate revenue model estimates spending per retained member. Observed revenue and contribution are preserved; only the remaining receipts are forecast, with recorded cost and refund relationships held fixed. Values cover each cohort's first twelve monthly receipts, discounted at 1% per month. The model has no calendar-time effect: common disruptions can move cohorts together and are outside these parameter bands. The main retention model extends Fader–Hardie's shifted-Beta-Geometric approach; the revenue model adapts Orduz's Gamma structure. An Orduz-inspired neural retention model is a separate validation benchmark, not the source of executive LTV. Parameter estimates and held-out checks appear in the appendix.

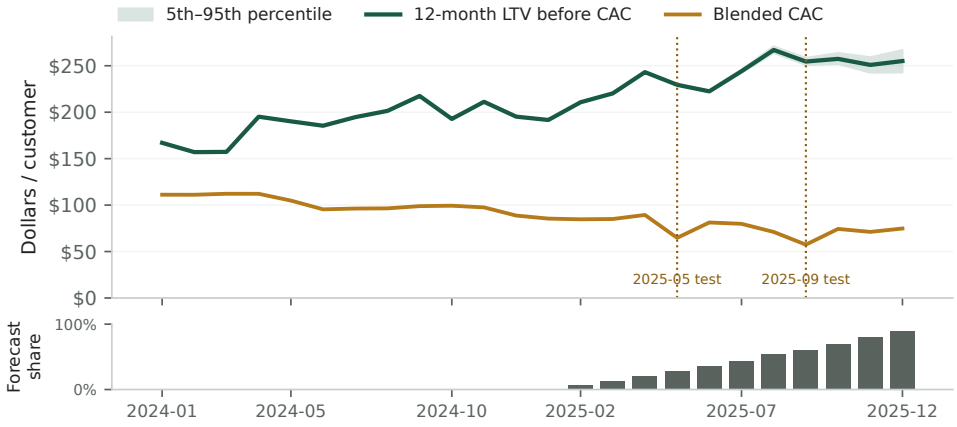


Figure 9. Observed and forecast. Common-horizon contribution LTV and blended CAC for all cohorts. The lower strip identifies the share of each LTV estimate that is forecast.

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval. The lower strip shows the forecast share of common 12-month LTV. Dotted markers identify experiment months with organic control buyers in the denominator. Blended CAC divides tracked spending by all acquired customers; it is not the causal cost of the next customer.

MANAGEMENT INTERPRETATION

Among cohorts with all twelve receipts observed, net contribution value increases from $55.83 for 2024-01 to $106.18 for 2025-01. The newest cohort, 2025-12, is separate: $180 net value, 90% forecast, with a 5th–95th parameter range of $168 to $192. Its projection depends on the cohort-trend term and customer mix, with no observed renewal trial. In the historical one-receipt comparison, the model over-forecast value by $20.15; that miss illustrates risk beyond the parameter band.

Future net revenue and contribution shares of payment revenue use observed channel-by-need-by-age shares, with the same group's observed renewal ages (age>=1) as fallback. Shares are fixed assumptions, not causally identified or assigned posterior uncertainty. Predicted customer value is not experimentally identified campaign return. A favorable average LTV-to-CAC comparison still requires a cash-recovery constraint and evidence that the next acquisition investment creates additional value.

Campaigns and management action

Incremental campaign evidence, investment conditions and accountable next steps.

Campaign incrementality and investment decisions

Randomized assignment includes all eligible prospects, including those who never purchase. Each test keeps its own control and follow-up window.

Table 4. Assigned prospects, buyers, recorded contribution and media spending by randomized arm.

Test	Arm	Assigned	Buyers	Contribution	Media
search / save time	control	250	25	$690.75	$0.00
search / save time	treatment	250	73	$1,817.45	$650.00
partner / build skill	control	250	29	$844.10	$0.00
partner / build skill	treatment	250	61	$1,659.15	$900.00

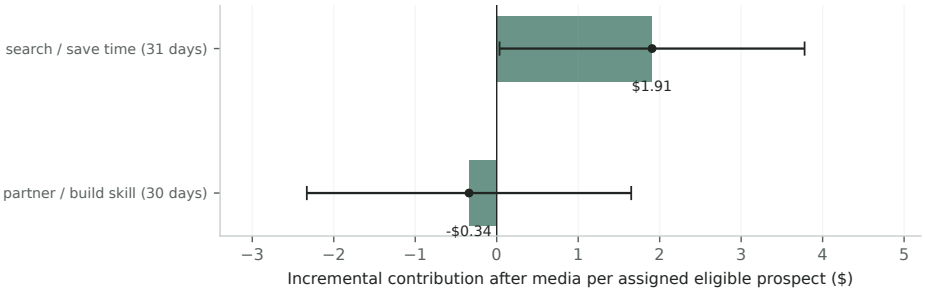


Figure 10. Randomized experiment. Incremental contribution after media per assigned prospect, with approximate 95% intervals over each test's recorded follow-up window.

Saved SQL outputs: `experiment_summary.csv` and `experiment_effects.csv`. Whiskers are approximate 95% intervals from all-assigned contribution variances, including nonbuyer zeros. Fixed media costs shift the mean, not its sampling variance. These 30/31-day windows do not measure lifetime campaign return. Later transactions exist, but the exports do not document whether controls remained protected after the prespecified follow-up.

THE SAME ECONOMICS APPLY TO TREATMENT AND CONTROL

For each arm, net revenue less recorded variable costs gives contribution. The difference in contribution per assigned eligible prospect is reduced by the treatment's additional media cost. Control purchases remain in the baseline rather than receiving campaign credit.

$$\Delta\text{contribution after media per prospect} = \frac{C_T - M_T}{N_T} - \frac{C_C - M_C}{N_C}.$$

MANAGEMENT ACTIONS AND ACCOUNTABILITY

Table 5. Accountable teams, next actions and the evidence required to revisit the recommendation.

Accountable team	Next action	Decision evidence
Marketing + Finance	Assess a larger budget using longer-window incremental contribution and cash recovery.	Age-aligned value and cash, alongside uncertainty in incremental returns.
Product + Operations	Investigate renewal changes using product events and customer feedback.	Renewal events, product actions and expressed customer evaluations.
Marketing + Analytics	Verify control protection before extending the comparison; otherwise run a longer test.	Contribution after media per assigned prospect, including nonbuyers, with uncertainty.
Data + Analytics	Refresh source exports and review forecast errors before the next decision.	SQL reconciliation, model holdouts, parameter output and file hashes.

Experimental conversion lift multiplied by predicted buyer LTV is an assumption-based projection, not an established lifetime campaign effect. The tested population, outcome window and valuation assumptions must remain visible.

Technical and data appendix

Model parameters, historical validation, source definitions and the reproducible analytical handoff.

Parameter meanings and fitted estimates

Technical appendix | Parameter definitions, units, priors and fitted uncertainty.

LIKELIHOOD, HETEROGENEITY AND ECONOMIC CONVERSION

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$p_{i,g} \sim \mathrm{Beta}(\alpha_g, \beta_g), \qquad \alpha_g = \mu_g \phi_g, \qquad \beta_g = (1-\mu_g)\phi_g,$$
$$N_{g,t} \mid N_{g,t-1} \sim \mathrm{Binomial}(N_{g,t-1}, 1-h_g(t)), \quad h_g(t) = \frac{\alpha_g}{\alpha_g+\beta_g+t-1}.$$

g indexes a cohort–need–channel cell; $t = 1$ is its first renewal. Search and save-time customers are the reference groups. Transformations, estimation settings and uncertainty scope are documented on the next page.

Table 6. Structural parameter estimates, priors, posterior intervals and sampling diagnostics.

Parameter	Prior	Mean	5th–95th	R-hat	ESS
cohort_scale	HalfNormal(0.25)	0.080	0.017 to 0.144	1.002	1,023
concentration_beta[intercept]	Normal(log(4),1)	1.120	0.977 to 1.273	0.999	2,621
concentration_beta[build_skill]	Normal(0,0.5)	0.388	0.153 to 0.626	1.001	2,965
concentration_beta[partner]	Normal(0,0.5)	0.119	-0.105 to 0.351	1.000	2,575
concentration_beta[organic]	Normal(0,0.5)	0.050	-0.660 to 0.807	1.000	5,677
mean_beta[intercept]	Normal(-2.5,1)	-1.336	-1.419 to -1.254	1.001	2,414
mean_beta[build_skill]	Normal(0,0.5)	-0.886	-0.996 to -0.773	1.001	3,108
mean_beta[partner]	Normal(0,0.5)	-0.243	-0.353 to -0.133	1.001	2,540
mean_beta[organic]	Normal(0,0.5)	-0.259	-0.692 to 0.154	1.000	5,272
mean_beta[cohort_trend]	Normal(0,0.5)	-0.356	-0.443 to -0.274	1.001	3,175
revenue_beta[intercept]	Normal(-0.5,0.8)	-0.607	-0.612 to -0.602	0.999	7,826
revenue_beta[log1p_age]	Normal(0.1,0.8)	0.106	0.103 to 0.108	0.999	5,893
revenue_beta[cohort_trend]	Normal(0,0.8)	0.028	0.024 to 0.032	1.000	7,768
revenue_beta[build_skill]	Normal(0,0.8)	0.063	0.059 to 0.068	1.000	9,064
revenue_beta[partner]	Normal(0,0.8)	0.005	0.001 to 0.009	0.999	8,936
revenue_beta[organic]	Normal(0,0.8)	0.053	0.027 to 0.078	1.000	6,434
revenue_shape	LogNormal(log(4),0.8)	16.216	15.160 to 17.311	0.999	6,758

The 24 standardized cohort offsets z have posterior means from -1.204 to 1.004; multiply each draw by cohort_scale to obtain its log-odds shift. Their largest R-hat is 1.001; smallest effective sample size is 3,315. The complete parameter_summary.csv provides every cohort effect, prior, interval and interpretation.

At the coefficient means, Search / save time at centered cohort trend and zero offset has mean latent monthly churn mu=20.8% and concentration phi=3.06. The cohort-trend coefficient's 5th–95th interval is -0.443 to -0.274; it is below zero, supporting lower mean churn in later cohorts within this model. This is a model-based description, not a causal effect of acquisition date.

Posterior means and 5th–95th percentiles are read from the saved, hash-bound parameter output. Units and reference groups are stated with the parameter definitions; transformed parameters are not interchangeable with their latent log-scale values.

Estimation settings and the scope of uncertainty

Technical appendix | Parameter precision, predictive accuracy and causal identification are different properties.

ESTIMATION AND REPRODUCIBILITY

Table 7. Recorded inference settings and convergence diagnostics for the full-data fit.

Setting or diagnostic	Recorded value
Inference	NUTS
Chains / warmup / samples	4 / 1000 / 1000 per chain
Posterior forecast draws	400
Maximum R-hat / minimum ESS	1.002 / 1,023
Divergences	0
Seed / numerical backend	20261116 / cpu / float64

FEATURE TRANSFORMATIONS AND INTERPRETATION

Mean-churn coefficients use a logit link; concentration and revenue use log links. Search and save-time customers are the reference groups. Retention cohort trend is $(\text{index} - 11.5)/11.5$; revenue cohort trend is $(\text{index} - 11.5)/12$. Revenue log-age is $\log(1 + \text{age})$. Cohort index runs from 0 to 23. A coefficient therefore needs its link, scale and reference group before it has a managerial interpretation.

INTERVAL SCOPE AND REMAINING RISK

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval.

One historical year-end origin, not proof of accuracy for every future cohort or business regime.

> Convergence diagnostics assess whether the fitted posterior was explored adequately. They do not establish that the model captures future customer behavior or identifies a campaign's causal effect. Historical holdouts test prediction using information available at the forecast origin.

METHOD SOURCES AND ADAPTATION

Retention extends Fader-Hardie sBG with mean/concentration parameterization, logit/log covariate links, a non-centered cohort effect on mean churn, grouped conditional Binomial renewal likelihood and NUTS estimation. Revenue adapts Orduz's Gamma structure but uses observed active counts in fitting, an additional per-member shape multiplier, a mean link without an age-by-cohort interaction, need/channel covariates and $100 scaling. Conditional on observed activity, the revenue and retention likelihoods have independent parameter blocks. The separate Orduz-inspired 4-hidden-unit Bayesian neural network is a retention comparison only; its cumulative-count likelihood is changed to conditional renewal trials.

The forecast component of customer contribution

Technical appendix | The same model forecasts retained customers, then converts that path into revenue and contribution.

FROM AN EXPECTED RENEWAL PATH TO CONTRIBUTION

Figure 2 introduces the extrapolation required for young cohorts. The revenue model applies expected spending per retained member to that renewal path. Recorded variable-cost and refund relationships translate revenue into contribution; acquisition cost is deducted once in net LTV.

OBSERVED EVIDENCE AND THE REMAINING FORECAST

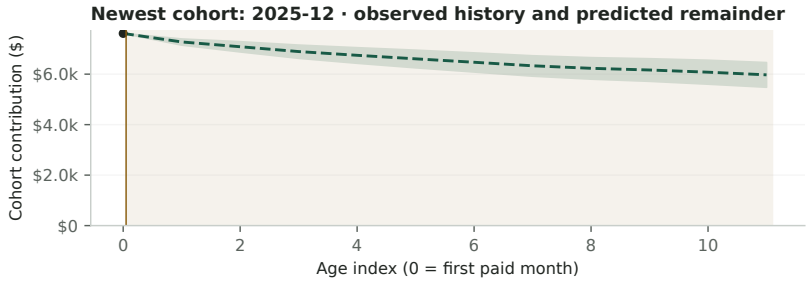


Figure 11. Observed and forecast. Observed contribution and the forecast remainder for the newest acquisition cohort.

Observed contribution is retained; only the unobserved remainder is predicted. The newest cohort's forecast relies on the cohort trend and customer mix until its own renewal history becomes available. Shading is the 5th–95th posterior parameter range. There is no calendar-time term; common shocks can affect multiple cohorts outside these bands.

These are expected customer paths conditional on fitted parameters. The bands exclude future customer-level Bernoulli variation, uncertain economic conversion and changes in the business regime. A narrow parameter band is not evidence that those risks are small.

Model fit and remaining timing patterns

Technical appendix | A model can describe observed counts closely and still miss the future.

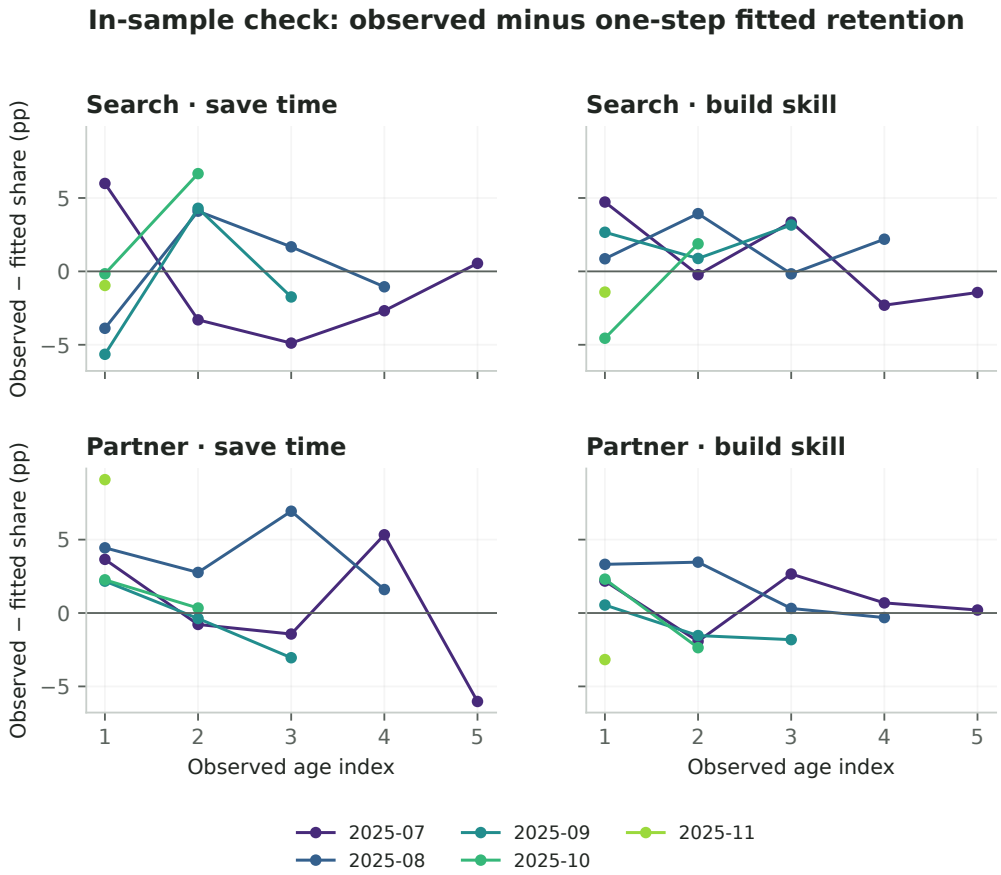


Figure 12. In-sample diagnostic. In-sample observed-minus-fitted retention residuals for the five newest cohorts with renewal history, by channel and need.

Saved SQL/model output: `model_fit_checks.csv`. These are in-sample, one-step fitted residuals conditional on previous observed customer counts. They are not held-out forecast errors.

WHERE CALENDAR-TIME FIT DEPARTS MOST

Table 8. The four largest absolute standardized calendar residuals among observed renewal risk sets.

Calendar month	Observed renewals	One-step fitted	Standardized residual
2025-04	1,632	1,665.4	-3.01
2025-03	1,555	1,584.4	-2.65
2024-10	1,057	1,034.5	+2.25
2025-08	2,222	2,199.1	+1.95

Four months with the largest absolute standardized calendar residual, selected from saved `calendar_fit_checks.csv`; only renewal ages 1 and older enter, not new acquisitions. These flag timing patterns, not causes, held-out errors or formal significance tests.

Historical prediction against withheld outcomes

Technical appendix | Forecasts are tested against outcomes withheld at historical forecast origins.

HELD-OUT PERFORMANCE

Training includes 2025-09; withheld months are 2025-10, 2025-11, 2025-12.

Withheld cells: 258. Retained-count mean absolute error: 1.22. Revenue weighted absolute percentage error: 5.9%. A separate training-only forecast checks complete 12-month values: LTV mean absolute error $4.02 versus $24.67 for its baseline. These tests evaluate prediction, not campaign effects.

Table 9. Prediction performance on historical holdouts, with explicit baseline comparisons and units.

Measure	Model	Baseline	Interpretation
Retained-count MAE	1.22	2.42	People per withheld cell
Retained-count WAPE	4.3%	Not reported	Retained customers, not revenue
Revenue WAPE	5.9%	12.8%	Withheld calendar months
12-month LTV MAE	$4.02	$24.67	Value per original customer

On the same 258 held-out cells, retained-count MAE is 1.215 for sBG, 1.335 for the Orduz-inspired 4-hidden-unit network, and 2.424 for the naive baseline. The network uses variational inference and a conditional monthly churn likelihood; it does not supply executive LTV. Revenue WAPE evaluates the main Gamma revenue model, not the retention-only network.

WHAT THE COMPARISON CAN ESTABLISH

The model and each baseline use the same withheld outcome cells. Their errors are reported in the stated units and at the same forecast horizon. A better retained-count forecast need not produce the better revenue forecast, and predictive performance does not establish a campaign's causal effect.

Outcomes after the 2026-01-01 cutoff do not enter estimation, tuning or features. Forecasts for the next twelve calendar months can be evaluated as those results become available. That window differs from the valuation of each cohort's first twelve monthly receipts.

Observed and forecast components of historical value

Technical appendix | Training includes the calendar month beginning 2024-12-01; later calendar months are used only for scoring.

At the 2024-12 forecast origin, 11 cohorts had an unfinished 12-month value horizon. Payback had already occurred for 7; those cases are excluded from forecast-month MAE. The 4 remaining cases with both observed and predicted recovery have payback-month MAE 0.00. This is a small future-payback sample, not evidence of universally exact prediction.

Table 10. Cohort-level historical value and payback forecasts, separating already observed recovery from forecast recovery.

Cohort	Observed / forecast months	Actual LTV	Model LTV	Baseline LTV	Payback actual / model	Payback evaluation
2024-02	11 / 1	$157.01	$157.15	$157.26	8 / 8	Already observed
2024-03	10 / 2	$157.27	$157.30	$157.48	8 / 8	Already observed
2024-04	9 / 3	$195.15	$194.34	$192.87	6 / 6	Already observed
2024-05	8 / 4	$190.16	$189.17	$185.80	6 / 6	Already observed
2024-06	7 / 5	$185.43	$183.94	$176.44	5 / 5	Already observed
2024-07	6 / 6	$194.67	$197.94	$184.43	5 / 5	Already observed
2024-08	5 / 7	$201.40	$192.77	$176.54	5 / 5	Already observed
2024-09	4 / 8	$217.49	$212.32	$184.41	5 / 5	Future; scored
2024-10	3 / 9	$192.74	$194.95	$143.43	5 / 5	Future; scored
2024-11	2 / 10	$211.19	$212.53	$156.92	5 / 5	Future; scored
2024-12	1 / 11	$195.21	$215.37	$111.72	5 / 5	Future; scored

Saved model output: `valuation_backtest.csv`. LTV values are discounted contribution before blended CAC, per original acquired customer. “Observed / forecast” counts the receipts within the common 12-month horizon; payback month counts receipts from 1.

WHY TOTAL-VALUE AND FORECAST-TAIL ERRORS ARE THE SAME

Observed contribution is copied identically into the actual and predicted totals. The forecast covers only the remaining months. Consequently, total LTV error equals the error in the unobserved tail; the long observed prefix does not create additional predictive accuracy. Rows are ordered by the amount of history available at the origin.

THE BASELINE AND ITS WEAKEST-INFORMATION CASE

The naive baseline freezes the last three observed renewal rates and revenue per retained customer. A cohort with only its acquisition month has no observed renewal rate; the baseline uses an explicit 0.8 renewal fallback. Its value is an assumption, not an estimated renewal rate. The December row makes this low-information comparison visible. The baseline underpredicts all nine rows with at least three forecast receipts; its tendency to underpredict increases the apparent advantage of the fitted model.

WHAT THE PAYBACK FORECAST REPRESENTS

Forecast payback is the first crossing of acquisition cost by the **posterior-mean expected contribution path**. It is not the mean or median of a payback-time distribution, and it does not estimate the chance that a realized customer cohort recovers its cost by that month. Posterior parameter bands also exclude future customer-level process noise.

> The small set of future-payback cases is a narrow diagnostic. Zero month error should be assessed alongside contribution-value errors and incomplete histories; it does not establish universal forecast accuracy. Outcomes after the observation cutoff are excluded from these historical checks.

Source systems and analytical definitions

Technical appendix | Eight separate exports become tested analytical views in DuckDB; no database server is required.

Eight source exports; 24 monthly acquisition cohorts; cutoff 2026-01-01. Pending renewal outcomes are excluded from observed renewal trials. All randomized prospects remain in experimental denominators, including nonbuyers.

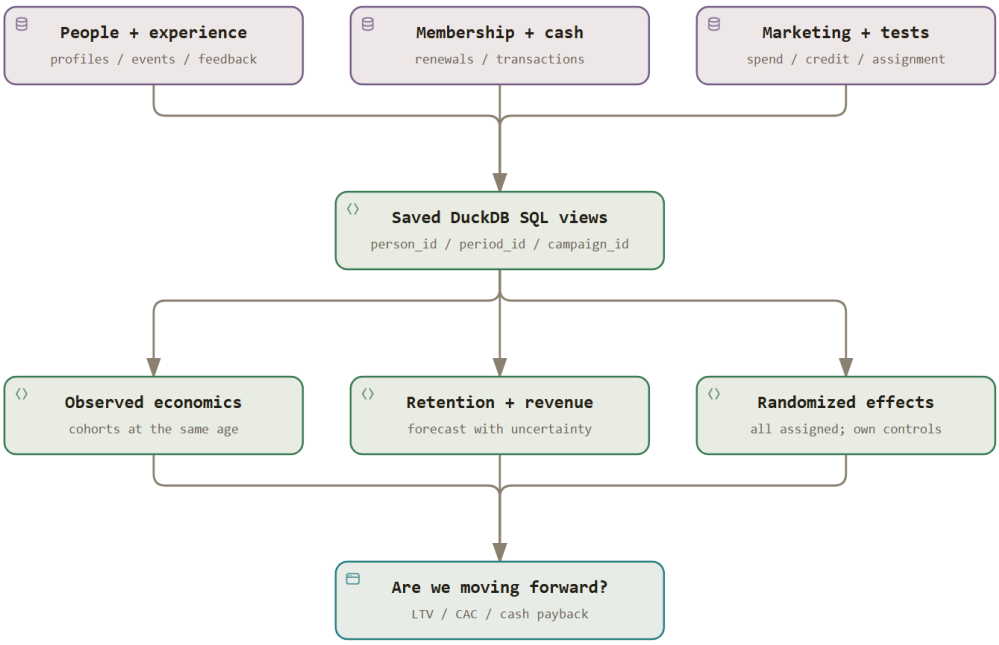


Figure 13. Eight source exports become analytical views and decision evidence through an auditable local pipeline.

The diagram is rendered by the bundled Archify package from original source specifications. The accompanying `diagrams/source-tables.html` and SVG show the full table-level schema and cardinality. Dataset classifications on the following pages list concrete fields.

KEYS DO NOT MAKE UNLIKE RECORDS INTERCHANGEABLE

A person can have many renewal periods, ledger transactions, product events and survey responses. Those tables must be aggregated at the intended grain before they are joined. Acquisition spending is a campaign-level record; it does not become a customer-level observation merely because it can be allocated across acquired customers. Experimental assignment defines an eligible population that includes noncustomers.

ANALYTICAL CONTRACTS

Table 11. Source-to-analysis definitions that preserve grain, timing, denominator and cost scope.

Definition	Contract
Joining cohort	Month of the first successful payment, derived from complete payment history.
Customer survival	Renewing contractual membership; original cohort size is the retained-share denominator.
Revenue and contribution	Ledger payments less discounts/refunds; contribution also subtracts recorded variable costs.
Acquisition investment	Tracked spend, including campaigns with no attributed acquisitions; no invented organic-source CAC.
Experimental unit	Every assigned eligible prospect, including nonbuyers and control buyers, within the prespecified follow-up window.
Unknown future	Right-censored ages stay unobserved. Missing survey responses remain missing perceptions.

Source hashes, saved SQL and downstream manifests bind the evidence chain. Numeric tables, editable LaTeX, source diagrams and the maintained style package accompany the report.

Source records, keys and cardinality

A3 landscape foldout | The eight raw exports remain separate. Campaign metadata is a derived lookup for a nonunique campaign-spend key.

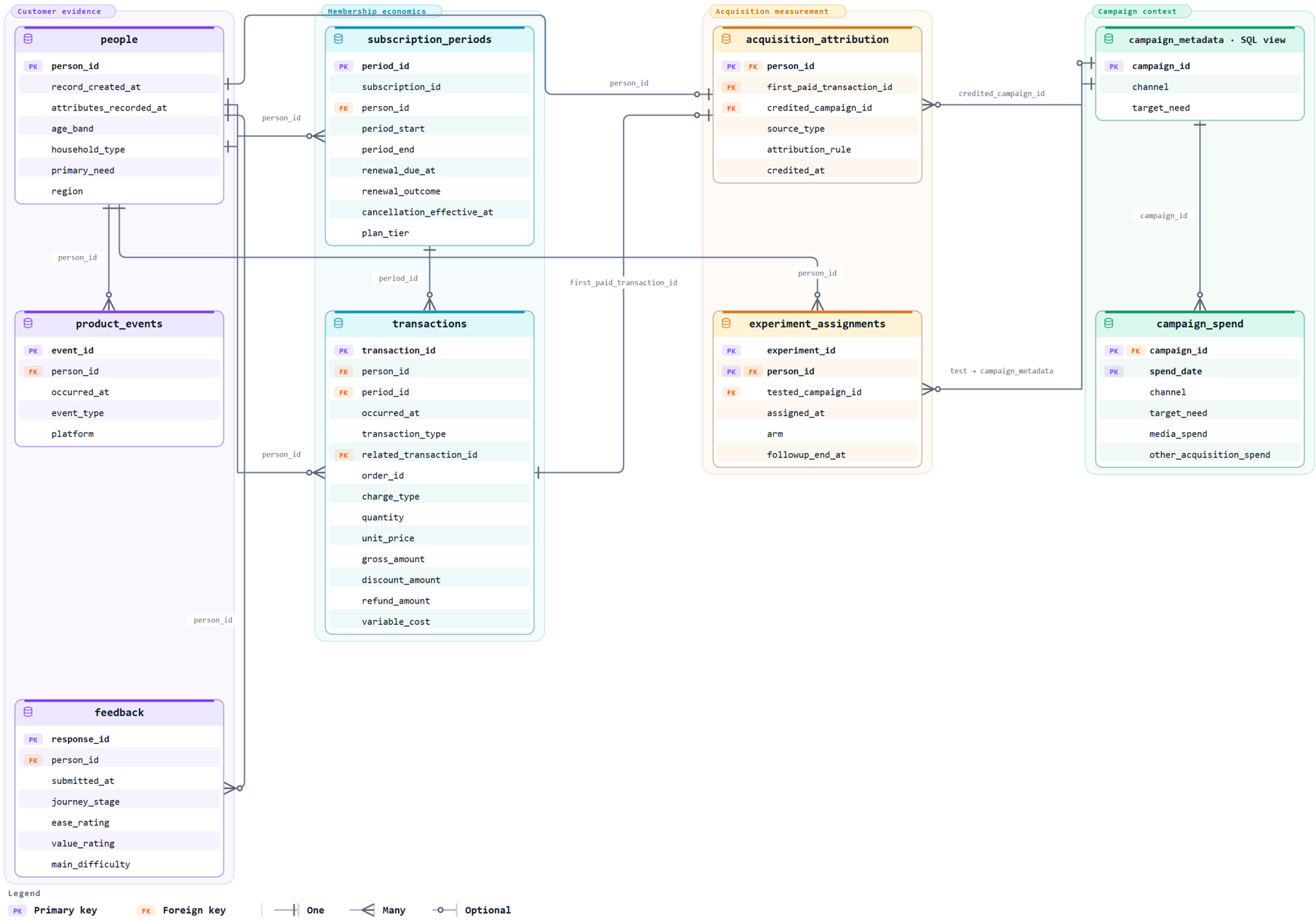


Figure 14. Source tables, join keys and cardinality, including the derived unique campaign lookup.

Native Archify schema; editable SVG/HTML: diagrams/source-tables. Refunds link transactions.related_transaction_id to transactions.transaction_id within the same ledger. Campaign foreign keys target unique campaign_metadata; campaign_spend uses campaign_id + spend_date. One-to-many sources are aggregated to the intended analytical grain before joining.

Source-level evidence in the Universal Customer View

Technical appendix | Context is marked explicitly; the categories describe evidence, not proof of an individual action or attitude.

Identifiers: Keys and profile descriptors identify or link people, groups and records; not every identifier is unique. Engagement: Observed customer interactions and their measurement context. Firm marketing activity and attribution metadata are retained as explicitly marked contextual evidence; they do not establish individual exposure, action or a causal effect. Relationship: Contractual and commercial arrangements, customer status, and explicitly marked operational context for relationship economics. Perception: Customer-reported views or intentions, distinguished from behavior and analyst inference.

Table 12. Universal Customer View: people.csv.

CRM | One person, including eligible prospects who never purchase.

Identifiers	Engagement	Relationship	Perception

`person_id`
`age_band`
`household_type`
`region`
	Not observed	`record_created_at`†
		`attributes_recorded_at`†
			`primary_need`

record_created_at: CRM relationship provenance; registration does not establish a paid customer. attributes_recorded_at: Profile provenance: verifies attributes were available before acquisition or assignment.

Table 13. Universal Customer View: subscription_periods.csv.

Renewal system | One scheduled monthly cycle; a paid outcome establishes retained membership for that cycle.

Identifiers	Engagement	Relationship	Perception

`period_id`
`subscription_id`
`person_id`
	Not observed	`period_start`
		`period_end`
		`renewal_due_at`
		`renewal_outcome`
		`cancellation_effective_at`
		`plan_tier`
			Not observed

Table 14. Universal Customer View: transactions.csv.

Finance | One payment or refund event; each payment is one order and refunds reuse its order ID.

Identifiers	Engagement	Relationship	Perception

`transaction_id`
`person_id`
`period_id`
`related_transaction_id`
`order_id`
	`occurred_at`
	`transaction_type`
	`charge_type`
	`quantity`
	`gross_amount`
	`discount_amount`
	`refund_amount`
		`unit_price`†
		`variable_cost`†
			Not observed

unit_price: The item's commercial price term is Relationship context; billed quantity and realized amounts record transaction Engagement. variable_cost: Firm-side accounting context for relationship economics; unlike billed amounts, it does not record a customer payment or reported perception.

Each raw field appears once in its dataset's classification. Source-specific rationale is preserved in `diagrams/ucv-tables.json`. A category marked not observed is a data gap, not an invitation to infer a missing attitude or relationship.

Customer activity, response and expressed perceptions

Technical appendix | Campaign spending and randomized assignment describe different objects from a customer's behavior or survey response.

Table 15. Universal Customer View: product_events.csv.

Product and support | One observed product/support interaction; no event does not mean no subscription.

Identifiers	Engagement	Relationship	Perception

`event_id`
`person_id`
	`occurred_at`
	`event_type`
	`platform`
		Not observed	Not observed

Table 16. Universal Customer View: feedback.csv.

Customer research | One completed linked survey response; nonresponse is missing, not a zero rating.

Identifiers	Engagement	Relationship	Perception

`response_id`
`person_id`
	`submitted_at`
	`journey_stage`†
		Not observed	`ease_rating`
			`value_rating`
			`main_difficulty`

journey_stage: Context of the response, not a measured universal customer path.

Table 17. Universal Customer View: campaign_spend.csv.

Marketing spending | One campaign per spending date; aggregate before joining to customer outcomes.

Identifiers	Engagement	Relationship	Perception
`campaign_id` `target_need`†	`spend_date`† `channel`† `media_spend`† `other_acquisition_spend`†	Not observed	Not observed

target_need: Identifies the campaign's target group; unlike a person's primary_need, it is not an individually stated intention. Channel describes the firm's marketing activity, so it is Engagement context instead. spend_date: Timing of firm marketing activity, not a customer touchpoint timestamp. channel: Context of firm marketing activity; it does not prove an individual's exposure. media_spend: Firm-side campaign spending is retained as Engagement context, not individual engagement or evidence of exposure. other_acquisition_spend: Firm-side creative/vendor costs are Engagement context, not observed customer activity; aggregate once before joining customers.

Marketing credit and experimental assignment

Technical appendix | A dagger marks contextual placement. It is not part of a source field's name.

Table 18. Universal Customer View: acquisition_attribution.csv.

Marketing attribution | One acquired person and the recorded credit for the first successful payment.

Identifiers	Engagement	Relationship	Perception
person_id first_paid_transaction_id credited_campaign_id	source_type† credited_at† attribution_rule†	Not observed	Not observed

source_type: Recorded acquisition-path credit; an attribution label does not establish incrementality. credited_at: When credit was recorded, not necessarily the acquisition or exposure time. attribution_rule: Measurement metadata for engagement credit, not a customer action itself.

Table 19. Universal Customer View: experiment_assignments.csv.

Experiment platform | One eligible person per randomized experiment, including all nonbuyers.

Identifiers	Engagement	Relationship	Perception
experiment_id person_id tested_campaign_id	Not observed	assigned_at† arm† followup_end_at†	Not observed

assigned_at: Experimental relationship/design metadata; assignment is not observed exposure or purchase. arm: Randomized intervention assignment; control nonbuyers must remain in the denominator. followup_end_at: Prespecified measurement window, not a subscription end date.

SOURCES AND MODEL ADAPTATIONS

Juan Orduz, Revenue Retention and Customer Lifetime Value with NumPyro; Peter Fader and Bruce Hardie, contractual-customer retention and shifted-Beta-Geometric models; Gil Bouhnick, Cohort Analysis. Exact links, adaptation boundaries and licensing are retained in the repository source documentation. Archify is pinned and redistributed with its license notices.

All graphics in this report are generated from this repository's original synthetic records or original diagram specifications. The archived Orduz source notebook is redistributed under its repository's MIT license with attribution and the license notice. Other third-party screenshots and private course materials are not included.

The explanatory value of baseline customer groups

Technical appendix | Identical customers and outcome windows make the four segmentation strategies comparable.

DISCRIMINATION ACROSS TWO-GROUP SEGMENTATION STRATEGIES

Among four two-group strategies on the same 4,278 paid customers, primary need has the lowest within-group variance in six-receipt contribution (4,303.5 dollars squared). It reduces unexplained variation by 8.3% relative to no segmentation; after centering outcomes within joining month, the reduction is 8.1%. Substantial individual variation remains within each group.

Table 20. Four strategies, each with two groups. Variance reductions use the corresponding unsegmented outcome variance as the denominator.

Two-group strategy	Within variance ($ squared)	Reduction	Cohort-centered reduction	Smallest group
acquisition channel	4,657.1	0.8%	0.7%	1,829
age band	4,692.2	0.0%	0.0%	2,110
primary need	4,303.5	8.3%	8.1%	2,002
region	4,693.6	0.0%	0.0%	2,069

Saved SQL outputs: `segmentation_profile.csv` and `segmentation_comparison.csv`. Every strategy uses the same mature paid customers and six-receipt horizon. Cohort centering subtracts each joining month's mean contribution before calculating within-group variance; it removes that month's shared outcome level without fitting a model.

WHAT THE VARIANCE COMPARISON ADDS

A segmentation can produce an actionable difference in average retention and dollars while leaving substantial variation among its members. The main report presents those economic differences first. This diagnostic asks how much individual contribution variation each two-group strategy accounts for; it does not establish a causal effect or predictive accuracy outside the observed sample.

> Cohort centering removes the mean level of each joining month. It does not adjust for every confounder. The channel-by-need four-cell comparison is presented separately and is not ranked against two-group strategies.

Knowledge assets and a reproducible handoff

Skills specify decisions and stage gates; reference assets explain methods; Python packages execute the transformations and models.

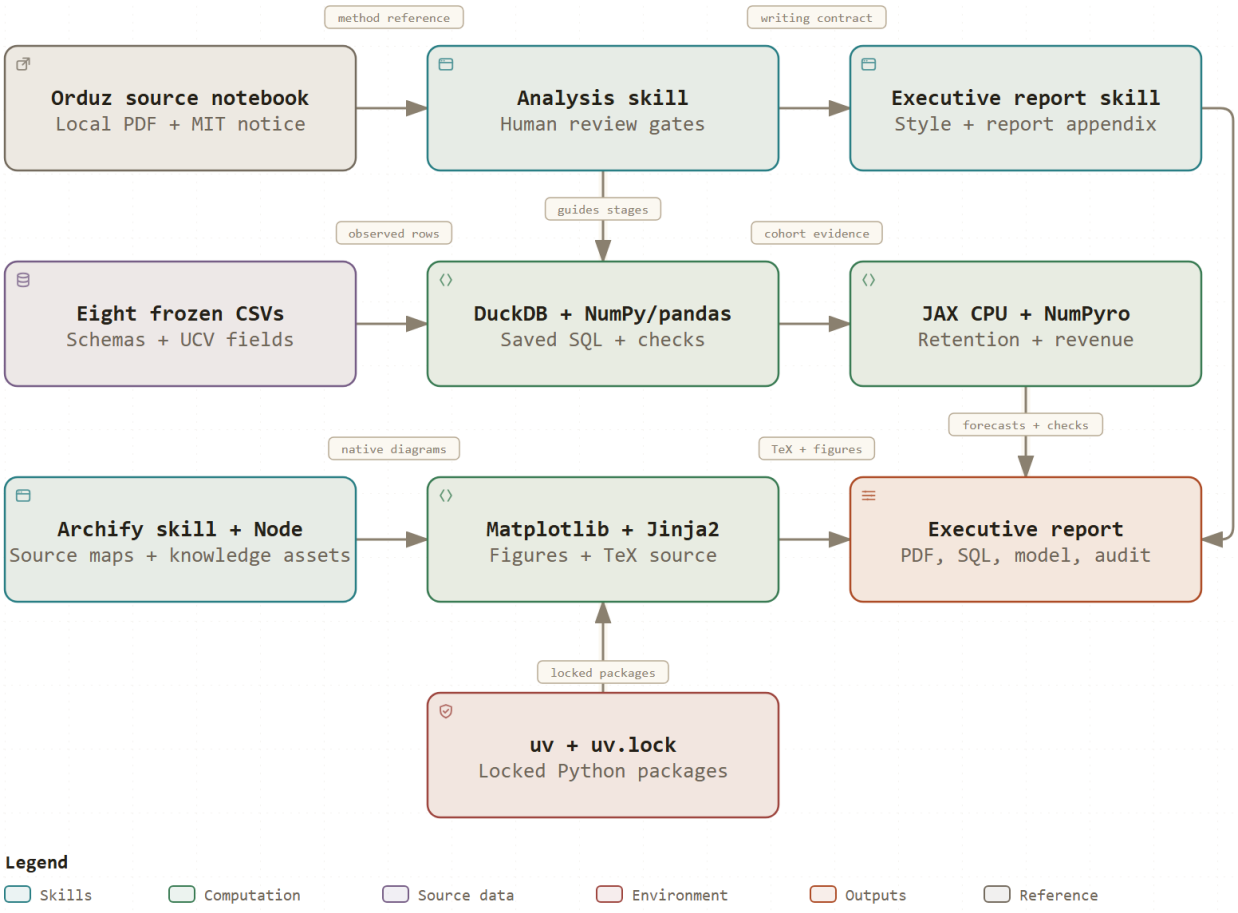


Figure 15. Analytical architecture. Knowledge assets and execution dependencies. Typed boxes, icons and colors distinguish skills, reference documents, Python packages, data and generated outputs.

THE SAME EVIDENCE SHOULD LEAD TO THE SAME REPORT

The recorded Python environment is installed with `uv sync --frozen`; commands run through `uv run --frozen`. The model uses CPU JAX/NumPyro; a GPU is not required. The analysis begins with the released records and does not regenerate them. Source hashes, saved SQL, model settings, figure data and document assets remain bound to the output manifests.

> Alternative models must use the same information available at the observation cutoff. Later outcomes are reserved for prospective evaluation and cannot inform features, parameter tuning or revisions to the input snapshot. Human review remains a gate between analytical stages.